Project: Worked examples
Title: Shaft under combined bending and torsion
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1. Scope and method

Calculates the bending and torsional stresses in a solid round shaft, with stress-concentration factors, and the factors of safety against yield. It covers static loading only, not fatigue or hollow shafts.

WHAT IT CALCULATES	RELATION
Bending stress, with its stress concentration	$\sigma = K_t \cdot 32M/(\pi d^3)$
Torsional shear stress, with its stress concentration	$\tau = K_{ts} \cdot 16T/(\pi d^3)$
Maximum shear stress (Tresca)	$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$
von Mises equivalent stress	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$
Factors of safety against yield, the lower governing	$n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$

Basis: Maximum-shear-stress and distortion-energy theories[1](§5-4 and §5-5, 10th ed.); Shaft design for stress[1](§7-4, 10th ed.).

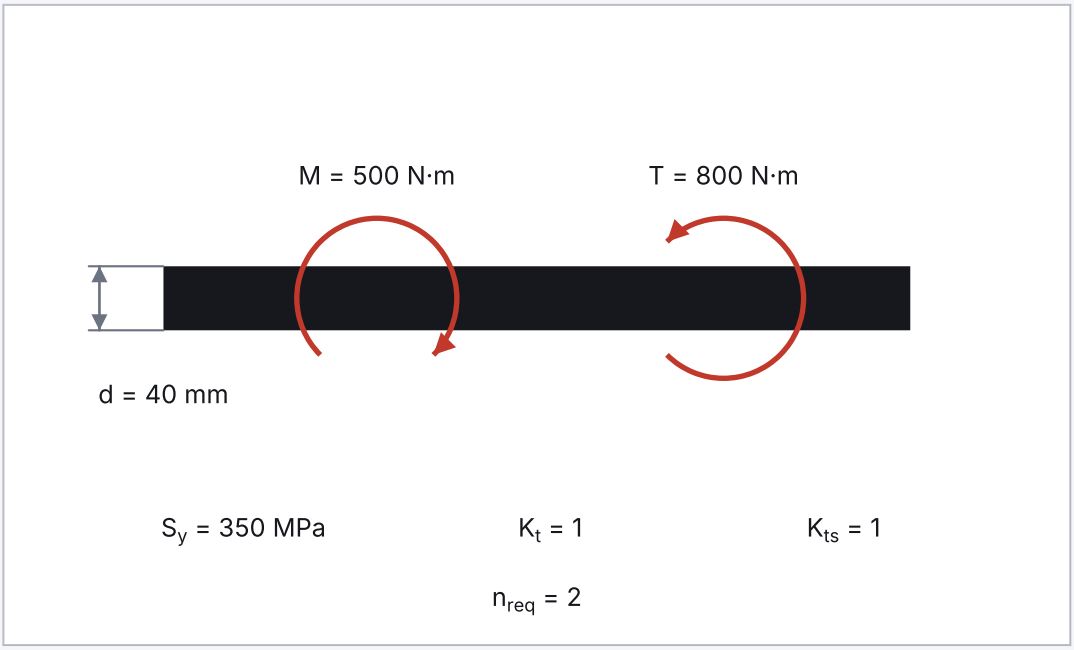


Figure 1 — A solid round shaft under bending and torsion at the section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	40	mm	drawing: DRG-SH-401 rev A	1 … 2000 mm
S_y	Yield strength	350	MPa	drawing: DRG-SH-401 rev A	50 … 3000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M	Bending moment (magnitude)	500	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
T	Torque (magnitude)	800	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
K_t	Bending stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
K_{ts}	Torsional stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
n_{req}	Required factor of safety	2		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. The shaft is a solid round section, linear elastic, under static loading. (The stress formulas of the method assume this.)

4. Working

[8] σ = K_t * 32 * M / (pi * d^3) [1](§7-4)
= 1 × 32 × 500 N·m / (π × (40 mm)^3)
= 79.577 MPa
Bending stress

[9] τ = K_{ts} * 16 * T / (pi * d^3) [1](§7-4)
= 1 × 16 × 800 N·m / (π × (40 mm)^3)
= 63.662 MPa
Torsional shear stress

[10] τ_{max} = sqrt((σ / 2)^2 + τ^2) [1](§5-4)
= sqrt((79.577 MPa / 2)^2 + (63.662 MPa)^2)
= 75.073 MPa
Maximum shear stress (Tresca)

[11] σ_{vm} = sqrt(σ^2 + 3 * τ^2) [1](§5-5)
= sqrt((79.577 MPa)^2 + 3 × (63.662 MPa)^2)
= 135.98 MPa
von Mises stress

[12] n_T = (S_y / 2) / τ_{max}
= 350 MPa / 2 / 75.073 MPa
= 2.3311
Factor of safety, maximum shear stress

[13] n_{VM} = S_y / σ_{vm}
= 350 MPa / 135.98 MPa
= 2.5739
Factor of safety, distortion energy

[14] n_{min} = min(n_T, n_{VM})
= min(2.3311, 2.5739)
= 2.3311
Governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION

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τ_{max}	75.07 MPa	Maximum shear stress
σ_{vm}	136.0 MPa	von Mises stress
n_{min}	2.331	Governing factor of safety
σ	79.58 MPa	Bending stress
τ	63.66 MPa	Torsional shear
n_T	2.331	Factor of safety (Tresca)
n_{VM}	2.5738	Factor of safety (von Mises)
u_n	0.86	Utilisation of factor of safety against the required factor (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	2	2.3311	0.86	[1](§7-4)

7. Applicability

None.

8. Verification

Template mech.shaft-stress@1.1.1 Content hash e68a699ed74d32f0 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.